Retail Sales Analytics and Performance Dashboard

1. Project Overview

This project analyzes retail sales data to understand sales performance, profitability, product category trends, regional performance, and customer purchasing patterns. The goal is to identify business opportunities, evaluate profitability, and provide actionable recommendations using Python, SQL, and Power BI.

2. Dataset Summary

- **Dataset:** Superstore Sales Dataset
- **Rows:** 9,994
- **Time Period:** 2011–2014
- **Key Features:**
    - Order information and order dates
    - Product categories and sub-categories
    - Sales, profit, quantity, and discount
    - Customer segments and regional information
    - Shipping details
- **Purpose:** Analyze retail performance and identify opportunities to improve profitability.

3. Exploratory Data Analysis Using Python

Python was used to explore and analyze the retail sales dataset.

- **Data Loading:** Imported the dataset using Pandas.
- **Initial Exploration:** Examined the dataset structure, data types, and summary statistics.
- **Data Preparation:** Reviewed data quality and prepared the dataset for analysis.
- **Category Analysis:** Compared sales and profit across product categories.
- **Regional Analysis:** Examined sales and profit across regions.
- **Trend Analysis:** Investigated monthly sales patterns.
- **Data Visualization:** Created charts to present sales and profitability patterns.

4. Data Analysis Using SQL

SQL was used to answer business questions and calculate key performance metrics.

1. **Overall Performance:** Calculated total sales, total profit, total orders, and quantity sold.
2. **Sales by Category:** Compared sales across Technology, Furniture, and Office Supplies.
3. **Profit by Category:** Identified the categories generating the highest profit.

4. **Regional Performance:** Compared sales and profit across regions.
5. **Monthly Sales Trends:** Examined sales performance over time.
6. **Customer Segment Analysis:** Compared sales across customer segments.
7. **Discount Analysis:** Investigated discount patterns and profitability.
8. **Annual Performance:** Compared sales across different years.
9. **Product Profitability:** Identified categories and products with relatively low profitability.

5. Dashboard in Power BI

An interactive Power BI dashboard was developed to present the results visually.

Dashboard components:

- KPI cards: Total Sales, Total Profit, Total Orders, and Total Quantity.
- Monthly Sales Trend.
- Sales by Category.
- Profit by Region.
- Sales by Customer Segment.
- Interactive slicers for Year, Region, and Category.
- Sales target achievement indicator using an assumed target of $2,500,000.

The dashboard allows users to filter the data and explore retail performance across different time periods, regions, and categories.

6. Key Performance Indicators and Findings

- **Total Sales:** $2,297,200.86
- **Total Profit:** $286,397.02
- **Total Orders:** 5,009
- **Total Quantity Sold:** 37,873

Key findings:

1. Technology recorded the highest category sales, at approximately $836,154.
2. Technology generated the highest category profit, at approximately $145,455.
3. The West region recorded the highest regional sales, at approximately $725,458.
4. Furniture generated approximately $18,451 in profit, considerably less than the other main categories.
5. Overall sales were approximately $2.30 million, with total profit of approximately $286,397.

7. Business Recommendations

- **Improve Furniture Profitability:** Review product costs, pricing, and discount strategies to identify opportunities to improve profit.

- **Regional Sales Strategy:** Study the West region's performance and identify opportunities to improve results in other regions.
- **Monitor Sales Trends:** Track monthly sales and profit to identify changes in performance.
- **Product Performance:** Focus on strong-performing categories while reviewing products with low profitability.
- **Data-Driven Decisions:** Use the interactive dashboard to compare categories, regions, and years.

8. Conclusion

This project demonstrates how Python, SQL, and Power BI can be used together to analyze retail sales data and support business decision-making. The analysis identifies sales trends, category performance, regional differences, and areas for potential profitability improvement. The interactive dashboard presents the findings in a clear and accessible format.

9. Screenshots

Dataset Exploration in Python

[3]:
```
print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset shape: (9994, 21)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sa
0	1	CA-2013-152156	09-11-2013	12-11-2013	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96
1	2	CA-2013-152156	09-11-2013	12-11-2013	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94
2	3	CA-2013-138688	13-06-2013	17-06-2013	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.62
3	4	US-2012-108966	11-10-2012	18-10-2012	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.57
4	5	US-2012-108966	11-10-2012	18-10-2012	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.36

5 rows × 21 columns

[6]:
```
print("Data types:")
display(df.dtypes)
```

```
Data types:
Row ID            int64
Order ID         object
Order Date       object
Ship Date        object
Ship Mode        object
Customer ID      object
Customer Name    object
Segment          object
Country          object
City             object
State            object
Postal Code       int64
Region           object
Product ID       object
Category         object
Sub-Category     object
Product Name     object
Sales           float64
Quantity          int64
Discount        float64
Profit          float64
dtype: object
```

[7]:
```
print("Number of rows and columns:", df.shape)
```

```
Number of rows and columns: (9994, 21)
```

[8]:
```
print("Exact duplicate rows:", df.duplicated().sum())
```

```
Exact duplicate rows: 0
```

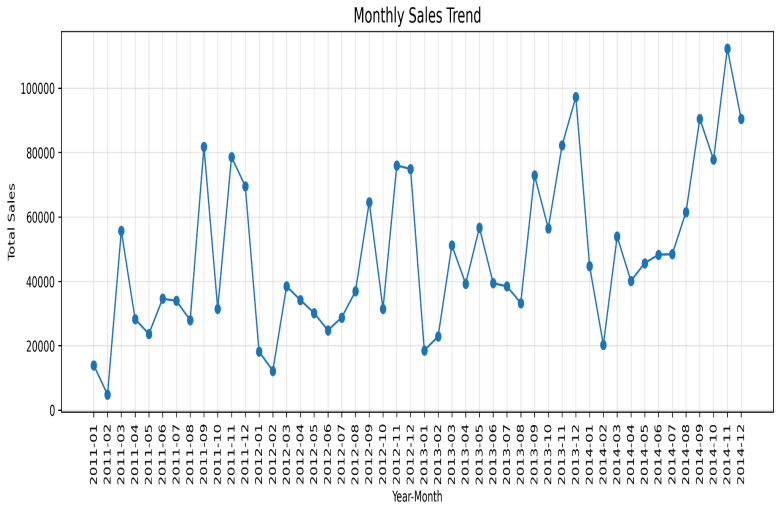
[12]:
```
df["Order Year"] = df["Order Date"].dt.year
df["Order Month"] = df["Order Date"].dt.month
df["Month Name"] = df["Order Date"].dt.strftime("%B")
df["Order Year-Month"] = df["Order Date"].dt.to_period("M").astype(str)

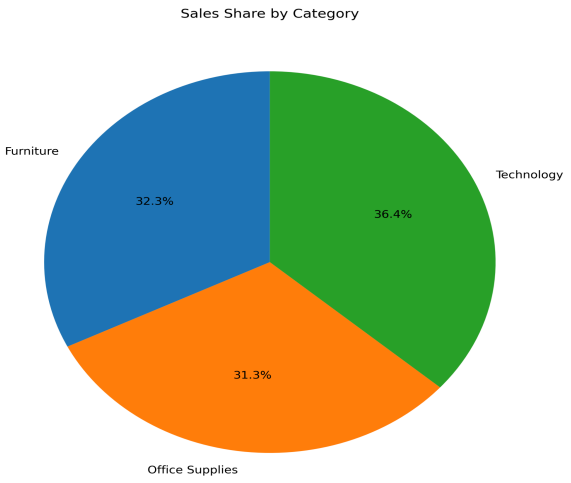
df["Shipping Days"] = (
    df["Ship Date"] - df["Order Date"]
).dt.days

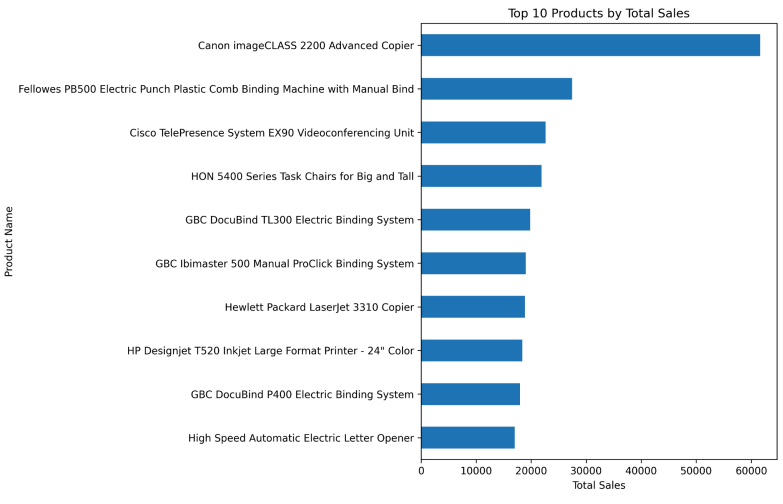
display(df[[
    "Order Date",
    "Order Year",
    "Month Name",
    "Order Year-Month",
    "Sales",
    "Profit",
    "Shipping Days"
]].head())
```

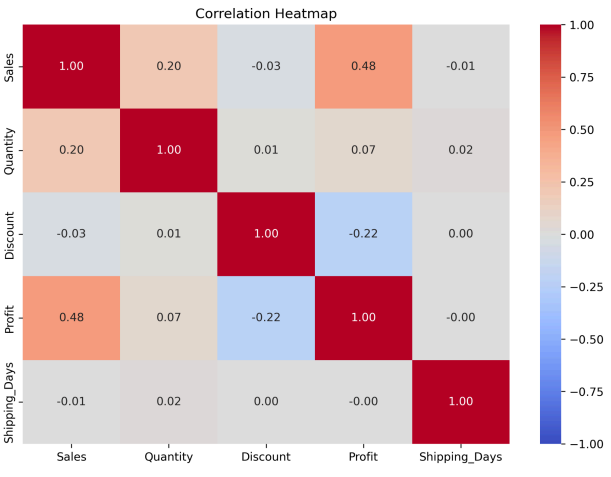
	Order Date	Order Year	Month Name	Order Year-Month	Sales	Profit	Shipping Days
0	2013-11-09	2013	November	2013-11	261.9600	41.9136	3
1	2013-11-09	2013	November	2013-11	731.9400	219.5820	3
2	2013-06-13	2013	June	2013-06	14.6200	6.8714	4
3	2012-10-11	2012	October	2012-10	957.5775	-383.0310	7
4	2012-10-11	2012	October	2012-10	22.3680	2.5164	7

Python Data Visualization









SQL Business Analysis

```sql
USE retail_sales_analytics;

# Q1. How is the business performing overall?
SELECT
    ROUND(SUM(Sales), 2) AS total_revenue,
    ROUND(SUM(Profit), 2) AS total_profit,
    COUNT(DISTINCT Order_ID) AS total_orders,
    SUM(Quantity) AS units_sold,
    ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS profit_margin_pct
FROM superstore;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_revenue	total_profit	total_orders	units_sold	profit_margin_pct
2297200.86	286397.02	5009	37873	12.47

```sql
# Q2. Which product categories generate the most revenue and profit?
SELECT
    Category,
    ROUND(SUM(Sales), 2) AS revenue,
    ROUND(SUM(Profit), 2) AS profit,
    ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS profit_margin_pct
FROM superstore
GROUP BY Category
ORDER BY revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Category	revenue	profit	profit_margin_pct
Technology	836154.03	145454.95	17.4
Furniture	741999.8	18451.27	2.49
Office Supplies	719047.03	122490.8	17.04

```sql
# Which regions perform best and worst?
SELECT
    Region,
    ROUND(SUM(Sales), 2) AS revenue,
    ROUND(SUM(Profit), 2) AS profit,
    COUNT(DISTINCT Order_ID) AS total_orders
FROM superstore
GROUP BY Region
ORDER BY profit DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

Region	revenue	profit	total_orders
West	725457.82	108418.45	1611
East	678781.24	91522.78	1401
South	391721.9	46749.43	822
Central	501239.89	39706.36	1175

Power BI Dashboard

